Macroeconomics in the Sequence Space

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Why Sequence Space?

Sequence Space vs. State Space

Motivation

- In heterogeneous-agent (HA) models, the economy's state is the entire cross-sectional distribution — an infinite-dimensional object.
- Sequence-space methods *sidestep* this challenge: they characterize local dynamics around the steady state without ever carrying the distribution as a state variable.
- The key insight: aggregate equilibrium responses to shocks can be encoded in Jacobian matrices of aggregate outcomes with respect to aggregate inputs.

Sequence Space vs. State Space

State Space

Definition 1

A state-space representation describes the economy recursively. At date t, a state variable x_t summarizes all payoff-relevant history, and the model evolves as

$$x_{t+1} = f(x_t, \varepsilon_{t+1}), \qquad y_t = g(x_t).$$

Given x_t, the future is independent of the past.

Definition 2

A sequence-space representation characterizes equilibrium as a mapping between entire time paths. We seek sequences $\{X_t\}_{t=0}^{\infty}$ such that equilibrium conditions hold at every date:

$$F(\{X_t\}_{t=0}^{\infty}, \{Z_t\}_{t=0}^{\infty}) = \mathbf{0},$$

where $\{X_t\}$ collects endogenous variables and $\{Z_t\}$ collects exogenous shocks or policy paths.

Sequence Space vs. State Space

Comparison

Intuition

- State space asks: *what is the current state, and how does it transition?*
- Sequence space asks: *given a shock path, what equilibrium outcome path clears all markets?*

Takeaway

When the distribution is the intractable state variable, sequence space reframes a "huge state" problem as a structured Jacobian problem — amenable to fast, scalable computation.

Local Solution

Example: The Krusell-Smith Model

Setup

- The economy is populated by a mass 1 of heterogeneous households,
- Each household maximizes its expected lifetime utility:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right],$$

- Standard CRRA utility,
- There exist n_e idiosyncratic states w/ transition matrix $P(e, e')$ and stationary distribution π,
- Agents supply an exogenous number of hours n, and earn wage income $w_t e n$,
- Agents can only trade in capital k, which pays a rental rate r_t net of depreciation.

Example: The Krusell-Smith Model

Household Problem

The value function of an agent entering period w/ state (e, k_-) is given by:

$$V_t(e, k_-) = \max_{c,\, k} u(c) + \beta \sum_{e'} V_{t+1}(e', k)\, P(e, e')$$

$$\text{s.t.} \qquad c + k = (1 + r_t)k_- + w_t e n$$

$$k \geq 0$$

- Denote by $c_t^*(e, k_-)$ and $k_t^*(e, k_-)$ the policy functions that solve the Bellman equation.
- Also denote by $D_t(e, K_-)$ the measure of households in state e that own capital in a set K_- at the start of date t.

Example: The Krusell-Smith Model

Household Problem

The distribution D_t has law of motion

$$D_{t+1}(e', K) = D_t(e, k_t^{*-1}(e, K))P(e, e').$$

where $k_t^{*-1}(e, K)$ denotes the inverse of $k_t^*(e, K)$.

- We assume that prior to $t = 0$, the economy is in a steady state.
- We suppose that agents start in this stationary distribution at date 0, so that $D_0 = D_{ss}$.

Example: The Krusell-Smith Model

Aggregate Capital

- For any t, the policy $k_t^*(e, k_-)$ is a function of the future path $\{r_s, w_s\}_{s=t}^{\infty}$.
- Given $D_0 = D_{ss}$, the distribution $D_t(e, K)$ at any t is a function of the entire path $\{r_s, w_s\}_{s=0}^{\infty}$.
- The aggregate household capital holdings are characterized by a capital function $\mathcal{K}_t(\{r_s, w_s\}_{s=0}^{\infty})$.

$$\mathcal{K}_t\left(\{r_s, w_s\}_{s=0}^{\infty}\right) = \sum_e \int_{k_-} k_t^*(e, k_-) D_t(e, dk_-)$$

Key Insight

The ability to reduce interactions between heterogeneous agents to functions such as $\mathcal{K}_t$, which map aggregate sequences into aggregate sequences, is key to the sequence-space Jacobian method.

Example: The Krusell-Smith Model

Production

- Firm adopts a Cobb-Douglas production function:

$$Y_t = Z_t K_t^{\alpha} N_t^{1-\alpha}$$

- The firm's first-order conditions:

$$r_t = \alpha Z_t K_{t-1}^{\alpha-1} N_t^{1-\alpha} - \delta$$
$$w_t = (1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}$$

Key Insight

The paths of prices $\{r_t, w_t\}_{t=0}^{\infty}$ are now functions of the exogenous paths $\{Z_t, N_t\}_{t=0}^{\infty}$ and the endogenous path $\{K_t\}_{t=0}^{\infty}$.

Example: The Krusell-Smith Model

Capital Market-Clearing Condition

Combining together, we can express the capital market-clearing condition at each point in time as a function H_t:

$$H_t(\boldsymbol{K}, \boldsymbol{Z}) \equiv \mathcal{K}_t \left(\left\{ \underbrace{\alpha Z_s \left(\frac{K_{s-1}}{\sum \pi(e) en} \right)^{\alpha-1} - \delta}_{r_s}, \underbrace{(1-\alpha) Z_s \left(\frac{K_{s-1}}{\sum \pi(e) en} \right)^{\alpha}}_{w_s} \right\}_{s=0}^{\infty} \right) - K_t = 0$$

where $\boldsymbol{K}$ is the vector of capital holdings $\{K_t\}_{t=0}^{\infty}$ and $\boldsymbol{Z}$ is the vector of exogenous shocks $\{Z_t\}_{t=0}^{\infty}$.

Equilibrium Capital Path

Given initial capital K_{-1} and the exogenous path for productivity $\boldsymbol{Z}$, we have equilibrium path of capital $\{K_t\}_{t=0}^{\infty}$ by solving the nonlinear equation $H_t(\boldsymbol{K}, \boldsymbol{Z}) = 0$.

Example: The Krusell-Smith Model

IRF

By *implicit function theorem*, given an exo. shock $\mathrm{d}\boldsymbol{Z} = (\mathrm{d}Z_0, \mathrm{d}Z_1 \ldots)'$ at $t = 0$, the linear impulse response of capital to this shock is given by:

$$d\boldsymbol{K} = -H_K^{-1} H_Z \, d\boldsymbol{Z}$$

where H_K and H_Z denote the Jacobians of H with respect to $\boldsymbol{K}$ and $\boldsymbol{Z}$, evaluated at the steady state.

- Given $\mathrm{d}\boldsymbol{K}$, the impulse responses of other variables—for example, $\{Y_t, r_t, w_t\}_{t=0}^{\infty}$—follow immediately.
- In practice, $H_t(\boldsymbol{K}, \boldsymbol{Z}) = 0$ is solved up to a sufficiently large horizon T such that $\boldsymbol{K}$ and $\boldsymbol{Z}$ have approximately returned to their steady states by time T.

Example: The Krusell-Smith Model

Constructing the Jacobian

- We use the *chain rule* to relate the Jacobians H_K and H_Z to the derivatives of the $\mathcal{K}$.
- For example, differentiating $H_t(\boldsymbol{K}, \boldsymbol{Z}) = 0$ with respect to K_s
- The (t, s) entry of H_K is:

$$[H_K]_{t,s} = \frac{\partial \mathcal{K}_t}{\partial r_{s+1}} \frac{\partial r_{s+1}}{\partial K_s} + \frac{\partial \mathcal{K}_t}{\partial w_{s+1}} \frac{\partial w_{s+1}}{\partial K_s} - \mathbf{1}_{\{s=t\}}$$

- A similar expression applies to H_Z.
- In addition, the derivatives such as $\frac{\partial r_{s+1}}{\partial K_s}$ at (K^{ss}, Z^{ss}) can all be computed analytically.
- For example:

$$\frac{\partial r_{s+1}}{\partial K_s} = \alpha(\alpha - 1) Z^{ss} (K^{ss})^{\alpha-2} (N^{ss})^{1-\alpha}$$

Example: The Krusell-Smith Model

Constructing the Jacobian

Recall that:

$$d\boldsymbol{K} = -H_K^{-1} H_Z \, d\boldsymbol{Z},$$

To get $H_K^{-1} H_Z$, the only thing left is to solve the Jacobians of the $\mathcal{K}$ function with respect to its two inputs r and w.

$$[H_K]_{t,s} = \frac{\partial \mathcal{K}_t}{\partial r_{s+1}} \frac{\partial r_{s+1}}{\partial K_s} + \frac{\partial \mathcal{K}_t}{\partial w_{s+1}} \frac{\partial w_{s+1}}{\partial K_s} - \mathbf{1}_{\{s=t\}}$$

Let

$$\mathcal{J}_{t,s}^{\mathcal{K},r} = \frac{\partial \mathcal{K}_t}{\partial r_s}, \quad \mathcal{J}_{t,s}^{\mathcal{K},w} = \frac{\partial \mathcal{K}_t}{\partial w_s}$$

be the Jacobians up to some truncation horizon T.

Example: The Krusell-Smith Model

Constructing the Jacobian

Sufficient statistics

The Jacobians $\mathcal{J}^{\mathcal{K},r}$ and $\mathcal{J}^{\mathcal{K},w}$ are sufficient statistics for the household in sequence space equilibrium:

- Shocks propagate through the dynamics of the household capital distribution, but *we do not need those details if we have the Jacobians.*
- Column s of $\mathcal{J}^{\mathcal{K},r}$ is the impulse response of capital to a news shock dr_s; the Jacobian gives these for all $s = 0, \ldots, T-1$.
- These are rich objects, but *there is a fast new algorithm to compute them.*

Fake News Algorithm
General Model Representation

We represent a heterogeneous-agent problem as a mapping from a time path of aggregate inputs $\{\boldsymbol{X}_t\}$ to a time path of aggregate outputs $\{\boldsymbol{Y}_t\}$.

Assume:

- $\boldsymbol{X}_t \in \mathbb{R}^{n_x}$: aggregate inputs,
- $\boldsymbol{Y}_t \in \mathbb{R}^{n_y}$: aggregate outputs,
- the distribution is discretized on n_g grid points,
- $\boldsymbol{D}_t \in \mathbb{R}^{n_g}$: cross-sectional distribution of agents at time t.

Aggregate outputs are averages of individual outcomes against the distribution:

$$\boldsymbol{Y}_t = \boldsymbol{y}_t' \boldsymbol{D}_t,$$

where $\boldsymbol{y}_t$ is an $n_g \times n_y$ matrix of individual outcomes (for example, consumption, savings, labor supply, or other individual-level objects).

Fake News Algorithm

Three Core Equations

We assume there exist three functions v, Λ, and y such that, for a given initial distribution $\boldsymbol{D}_0$, the aggregate sequence solves

$$\boldsymbol{v}_t = v(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t),$$

$$\boldsymbol{D}_{t+1} = \Lambda(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)\, \boldsymbol{D}_t,$$

$$\boldsymbol{Y}_t = y(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)' \boldsymbol{D}_t.$$

Interpretation:

- $\boldsymbol{v}_t$: backward-looking object, typically the value function;
- $\Lambda(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)$: transition matrix governing the law of motion of the distribution;
- $y(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)$: individual outcomes used to aggregate into $\boldsymbol{Y}_t$.

Fake News Algorithm

Economic Meaning of the System

Step 1: Backward

Given the future object $\boldsymbol{v}_{t+1}$ and current aggregates $\boldsymbol{X}_t$, solve for the current object $\boldsymbol{v}_t$.

Step 2: Forward

Use $\Lambda(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)$ to update the cross-sectional distribution:

$$\boldsymbol{D}_{t+1} = \Lambda(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)\boldsymbol{D}_t.$$

Step 3: Aggregate

Aggregate individual outcomes across agents:

$$\boldsymbol{Y}_t = y(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)'\boldsymbol{D}_t.$$

Fake News Algorithm

Steady State and Transition Paths

For a constant aggregate input $\boldsymbol{X}_{ss}$, the steady state is a fixed point

$$(\boldsymbol{Y}_{ss}, \boldsymbol{v}_{ss}, \boldsymbol{D}_{ss})$$

of the system.

For convenience, define

$$\Lambda_{ss} \equiv \Lambda(\boldsymbol{v}_{ss}, \boldsymbol{X}_{ss}), \qquad y_{ss} \equiv y(\boldsymbol{v}_{ss}, \boldsymbol{X}_{ss}).$$

We consider transition paths of length T that end at the steady state:

$$\boldsymbol{X}_{T-1} = \boldsymbol{X}_{ss}, \qquad \boldsymbol{v}_T = \boldsymbol{v}_{ss}.$$

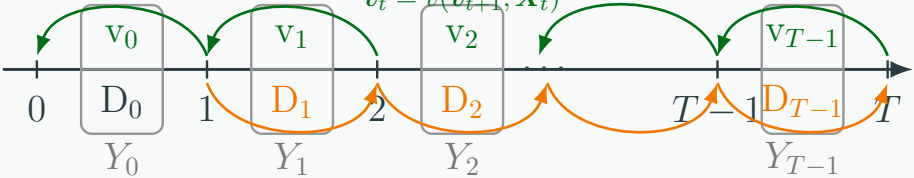
The initial distribution $\boldsymbol{D}_0$ is given and often set equal to $\boldsymbol{D}_{ss}$. Hence, this framework is designed to study transitory shocks around a steady state.

Fake News Algorithm

Three-Step Structure of the HA Model

Step 1: Backward solve

$$\boldsymbol{v}_t = v(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)$$



Step 2: Forward update

$$\boldsymbol{D}_{t+1} = \Lambda(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)\boldsymbol{D}_t$$

Step 3: Aggregate outcomes

$$\boldsymbol{Y}_t = y(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)'\boldsymbol{D}_t$$

Determinacy and Large-Scale Solutions in the Sequence Space

To be developed

2nd SSJ

To be developed

SSJ in Continuous Time

To be developed

Applications of the extended version of SSJ

- Fiscal and monetary policy [Auclert et al., 2025c, Auclert et al., 2024b]
- Het households + het firms [Winberry et al., 2025]
- Behavioural macro [Auclert et al., 2020, Angeletos et al., 2025]
- Life-cycle model
  [Bardóczy and Velásquez-Giraldo, 2024, Bardoczy and Velasquez Giraldo, 2025]
- Dynamic spatial model [Donald et al., 2025, Fan et al., 2023]
- Production network [Schaab and Tan, 2023]
- Het banks [Bellifemine et al., 2022]
- Optimal policy [Auclert et al., 2024a, McKay and Wolf, 2022]
- Macro labor [Guerreiro et al., 2024, Kekre, 2023]

Global Solution

Repeated transition method

To be developed

DL in sequence space

To be developed

RL in sequence space

To be developed

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